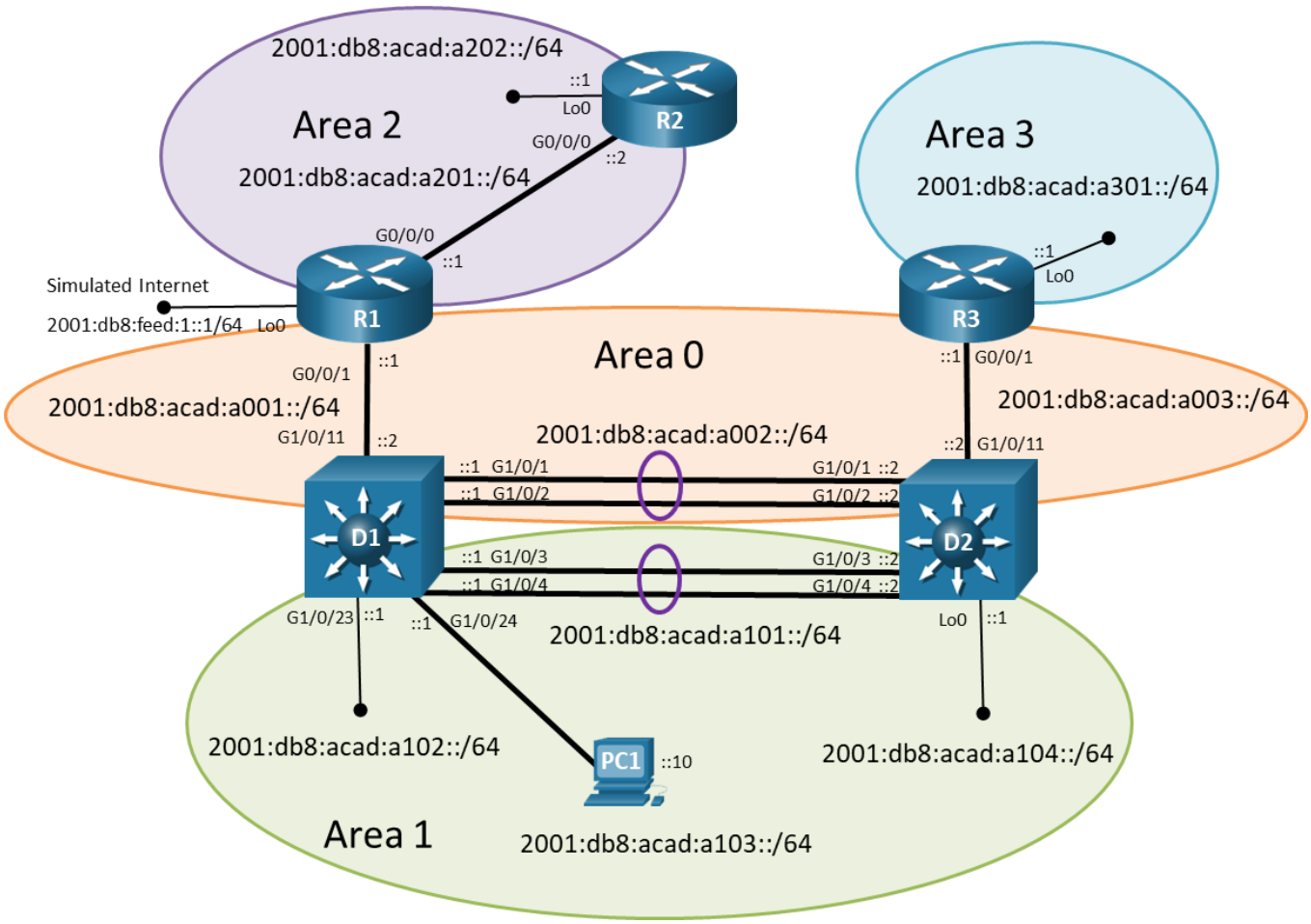
Lab - Troubleshoot OSPFv3

Topology



Addressing Table

Device	Interface	IPv6 Address	Link-Local
R1	G0/0/0	2001:db8:acad:a201::1/64	fe80::a201:1
	G0/0/1	2001:db8:acad:a001::1/64	fe80::a001:1
	Lo1	2001:db8:feed:1::1/64	fe80::1:1
R2	G0/0/0	2001:db8:acad:a201::2/64	fe80::a201:2
	Lo1	2001:db8:acad:a202::1/64	fe80::a202:1
R3	G0/0/1	2001:db8:acad:a003::1/64	fe80::a003:1
	Lo1	2001:db8:acad:a301::1/64	fe80::a301:1

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Device	Interface	IPv6 Address	Link-Local
D1	G1/0/1 (Po1)	2001:db8:acad:a002::1/64	fe80::a002:1
	G1/0/2 (Po1)	2001:db8:acad:a002::1/64	fe80::a002:1
	G1/0/3 (Po2)	2001:db8:acad:a101::1/64	fe80::a101:1
	G1/0/4 (Po2)	2001:db8:acad:a101::1/64	fe80::a101:1
	G1/0/5	2001:db8:acad:a103::1/64	fe80::a103:1
	G1/0/11	2001:db8:acad:a001::2/64	fe80::a001:2
	G1/0/23	2001:db8:acad:a102::1/64	fe80::a102:1
D2	G1/0/1 (Po1)	2001:db8:acad:a002::2/64	fe80::a002:2
	G1/0/2 (Po1)	2001:db8:acad:a002::2/64	fe80::a002:2
	G1/0/3 (Po2)	2001:db8:acad:a101::2/64	fe80::a101:2
	G1/0/4 (Po2)	2001:db8:acad:a101::2/64	fe80::a101:2
	G1/0/11	2001:db8:acad:a003::2/64	fe80::a003:2
	Lo0	2001:db8:acad:a104::1/64	fe80::a104:1
PC1	N/A	SLAAC	SLAAC

Note: To make it easier to recognize IPv6 prefixes, familiarize yourself with the IPv6 GUA and LLA address formats.

- **GUA**: The GUA has a 16-bit subnet-ID, a<area-id>xx ("a" for area). For example, subnet-ID a201 is area 2, network 01.
- **LLA**: Following best practice the LLA address is unique on each interface. The LLA interface-ID uses the GUA subnet-ID:interface-ID for the last 64 bits. For example, fe80::a201:1 has an LLA interface-ID a201 (the subnet-ID of the GUA) and :1 (the Interface ID of the GUA).

Objectives

Troubleshoot network issues related to the configuration and operation of OSPFv3.

Background / Scenario

Although the topology has a limited number of routers, you should use the appropriate troubleshooting commands to help find and solve the problems in the three trouble tickets as if this were a much more complex topology with many more routers and networks.

You will be loading configurations with intentional errors onto the network. Your tasks are to FIND the error(s), document your findings and the command(s) or method(s) used to fix them, FIX the issue(s) presented here and then test the network to ensure both of the following conditions are met:

1) the complaint received in the ticket is resolved
2) full reachability is restored

Note: The routers used with CCNP hands-on labs are Cisco 8200L with Cisco IOS XE Release 17.12.5 (universalk9 image). The switches used in the labs are Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.5 (universalk9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is

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shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Make sure that the devices have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 3 Routers (Cisco 8200L with Cisco IOS XE Release 17.12.5c universal image or comparable)
- 2 Switches (Cisco 9300L with Cisco IOS XE Release 17.12.5 universal image or comparable)
- 1 PC (Choice of operating system with terminal emulation program installed)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Trouble Ticket 9.1.1.1

Scenario:

Your university has just completed a phased migration to IPv6-only across the campus core and distribution layers, standardizing on multiarea OSPFv3 address families for routing. Different teams owned different parts of the rollout, and a mix of device models and IOS XE releases were involved. During post-migration validation, the NOC noticed that the 2001:db8:acad:a202::/64 prefix (R2's loopback in area 2) is missing from all router routing tables. Application teams also report failed reachability tests to 2001:db8:acad:a202::2. You have been asked to verify the OSPFv3 AF configuration consistency and ensure that the a202 prefix is properly learned by all routers without changing the existing area design or addressing.

Be sure you are using devices that have been connected and prepared by your instructor. Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/9.1.1.1-r1-config.txt running-config`
R2	`copy flash:/enarsi/9.1.1.1-r2-config.txt running-config`
R3	`copy flash:/enarsi/9.1.1.1-r3-config.txt running-config`
D1	`copy flash:/enarsi/9.1.1.1-d1-config.txt running-config`
D2	`copy flash:/enarsi/9.1.1.1-d2-config.txt running-config`

- All routers should have the 2001:db8:acad:a202::/64 network in their routing tables.
- All devices should be able to ping 2001:db8:acad:a202::1/64.
- When you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #
- Save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command. This script will clear your configurations and reload the devices.

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Part 2: Trouble Ticket 9.1.1.2

Scenario:

Following a scheduled maintenance window, several devices, including R3, were upgraded to the latest IOS XE and had their configurations restored from backups. After the upgrades, users in academic labs began reporting loss of access to the IPv6 resources in area 3. The NOC confirms that 2001:db8:acad:a301::/64 is not appearing in campus routing tables and pings to 2001:db8:acad:a301::1 are failing. Because some devices were rebuilt without legacy IPv4 management addresses, you are asked to review OSPFv3 address-family settings on R3 and restore end-to-end reachability for area 3, keeping the current area structure intact.

Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/9.1.1.2-r1-config.txt running-config`
R2	`copy flash:/enarsi/9.1.1.2-r2-config.txt running-config`
R3	`copy flash:/enarsi/9.1.1.2-r3-config.txt running-config`
D1	`copy flash:/enarsi/9.1.1.2-d1-config.txt running-config`
D2	`copy flash:/enarsi/9.1.1.2-d2-config.txt running-config`

- All routers should have the 2001:db8:acad:a301::/64 network in their routing tables.
- All devices should be able to ping 2001:db8:acad:a301::1/64.
- When you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #
- Then save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command. This script will clear your configurations and reload the devices.

Part 3: Trouble Ticket 9.1.1.3

Scenario:

A new router (R2) was recently added to area 2 to improve resilience and reduce failover times. During the cutover, engineers tuned OSPFv3 timers for faster detection, but reports from the field indicate that neighbors in area 2 are not fully forming and routes for 2001:db8:acad:a202::/64 are not propagating to the rest of the network. At the same time, there is a planned outage on R3's uplink to simulate a fiber cut, so traffic must still reach 2001:db8:acad:a301::1 via the remaining paths. Your task is to validate and correct the OSPFv3 timer settings between adjacent routers so that the a202 prefix is learned network-wide, while ensuring the 2001:db8:acad:a301::/64 network remains isolated in its own area as per design.

Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/9.1.1.3-r1-config.txt running-config`
R2	`copy flash:/enarsi/9.1.1.3-r2-config.txt running-config`
R3	`copy flash:/enarsi/9.1.1.3-r3-config.txt running-config`
D1	`copy flash:/enarsi/9.1.1.3-d1-config.txt running-config`

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D2	`copy flash:/enarsi/9.1.1.3-d2-config.txt running-config`

- PC1 should be able to ping 2001:db8:acad:a202::1.
- For each device, the adjacency state should be FULL with all neighbors.
- When you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #
- Save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command. This script will clear your configurations and reload the devices.

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